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Short-length Adversarial Training Helps LLMs Defend Long-length Jailbreak Attacks: Theoretical and Empirical Evidence

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Abstract

Jailbreak attacks against large language models (LLMs) aim to induce harmful behaviors in LLMs through carefully crafted adversarial prompts. To mitigate attacks, one way is to perform adversarial training (AT)-based alignment, *i.e.*, training LLMs on some of the most adversarial prompts to help them learn how to behave safely under attacks. During AT, the length of adversarial prompts plays a critical role in the robustness of aligned LLMs. While long-length adversarial prompts during AT might lead to strong LLM robustness, their synthesis however is very resource-consuming, which may limit the application of LLM AT. This paper focuses on adversarial suffix jailbreak attacks and unveils that to defend against a jailbreak attack with an adversarial suffix of length $\Theta(M)$, it is enough to align LLMs on prompts with adversarial suffixes of length $\Theta(\sqrt{M})$. Theoretically, we analyze the adversarial in-context learning of linear transformers on linear regression tasks and prove a robust generalization bound for trained transformers. The bound depends on the term $\Theta(\sqrt{M_{\text{test}}}/M_{\text{train}})$, where M_{train} and M_{test} are the numbers of adversarially perturbed in-context samples during training and testing. Empirically, we conduct AT on popular open-source LLMs and evaluate their robustness against jailbreak attacks of different adversarial suffix lengths. Results confirm a positive correlation between the attack success rate and the ratio of the square root of the adversarial suffix length during jailbreaking to the length during AT. Our findings show that it is practical to defend against "long-length" jailbreak attacks via efficient "short-length" AT. The code is available at https://github.com/fshp971/adv-icl.

1 Introduction

Large language models (LLMs) [5, 51, 28, 65] are widely adopted in various real-world applications to assist human users [55, 67, 57, 56, 24], but their safety is found to be vulnerable toward jailbreak attacks [60]. With carefully crafted adversarial prompts, one can "jailbreak" the safety mechanism of LLMs and induce arbitrary harmful behaviors [73, 7, 30]. To tackle the challenge, recent studies [63, 36, 68, 6] propose performing safety alignment through adversarial training (AT) [32] to enhance LLMs' robustness against jailbreaking. A standard AT for LLMs would train them on jailbreak prompts synthesized by strong jailbreak attacks to learn to refuse these harmful instructions [36].

In such AT, the length of synthesized adversarial prompts used for model training is critical to the final jailbreak robustness of LLMs. [3] and [64] have shown that longer adversarial prompts enjoy stronger

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jailbreaking abilities. Thus, it is reasonable to deduce that performing AT with longer adversarial prompts can help LLMs achieve stronger robustness to defend against "long-length" jailbreak attacks. However, synthesizing long-length adversarial prompts in adversarial training is resource-consuming since it requires solving discrete optimization problems in high-dimensional spaces, which thus needs lots of GPU memory and training time. This may limit the application of AT in LLMs' safety alignment and further raises the following research question: ***How will the adversarial prompt length during AT affect trained LLMs' robustness against jailbreaking with different prompt lengths?***

We study this research question by analyzing *suffix jailbreak attacks*, where each jailbreak prompt is constructed by concatenating a harmful instruction with a synthesized adversarial suffix. Our main finding is: **To defend against a suffix jailbreak attack with suffix length of $\Theta(M)$, it is enough to adversarially train LLMs on adversarial prompts with suffix length of only $\Theta(\sqrt{M})$.** In other words, we show that it is possible to defend long-length jailbreak attacks via efficient short-length AT.

Our finding is supported by *theoretical* and *empirical* evidence. Theoretically, we leverage the *in-context learning theory* [53, 69] to investigate how linear transformers learn linear regression tasks from in-context task samples under AT. To better simulate suffix jailbreak attacks in real-world LLMs, our analysis introduces a new *in-context adversarial attack*. Concretely, for any in-context task sample, this attack will adversarially perturb the last several in-context training points to maximize the squared prediction error that linear transformers made on the in-context test point. Under our theoretical framework, we prove a robust generalization bound for adversarially trained linear transformers. This bound has a positive correlation with the term $\Theta(\sqrt{M_{\text{test}}}/M_{\text{train}})$, where M_{train} and M_{test} are the number of perturbed in-context points in training and testing in-context task samples, respectively.

Empirically, we conduct AT with the GCG attack [73], one of the most effective jailbreak attacks, under various adversarial suffix lengths on five popular real-world LLMs and evaluate their robustness against jailbreak attacks with different adversarial suffix lengths. We use the jailbreak attack success rate (ASR) to express the robust generalization error of trained LLMs and find that this ASR has a clear positive correlation with the ratio of the square root of test-time adversarial suffix length to the AT adversarial suffix length. Such a correlation empirically verifies our main finding. We also find that AT with an adversarial suffix (token) length of 20 is already able to reduce the ASR of jailbreak attacks with an adversarial suffix (token) length of up to 120 by at least 30% in all experiments.

2 Related works

Jailbreak attacks. Jailbreaking [60] can be seen as adversarial attacks [49, 14] toward LLMs, which aim to synthesize adversarial prompts to induce targeted harmful behaviors from LLMs. Many efforts have been made on token-level jailbreak attacks, *i.e.*, searching adversarial prompts in the token space of LLMs, which can be achieved via gradient-based optimization [48, 16, 73, 26, 45, 71], heuristic greedy search [44, 17, 22], or fine-tuning prompt generators from pre-trained LLMs [38]. Other attempts include word-level adversarial prompt searching [30] or directly prompting LLMs to generate adversarial prompts [7, 29]. Our work focuses on token-level jailbreaking since it make it easier for us to control the adversarial prompt length for our analysis. More recent studies have found that increasing the length of adversarial prompts by adding more harmful demonstrations [3, 54, 61] or synthesizing longer adversarial suffixes [64] can make jailbreaking more effective. These works motivate us to investigate the problem of defending against "long-length" jailbreak attacks.

Adversarial training on LLMs. To defend against jailbreak attacks, a large body of studies focus on aligning LLMs to refuse responding jailbreak prompts [37, 42, 40, 41, 8]. More recent works have started to adopt adversarial training (AT) [32] to align LLMs. [36] trained LLMs on (discrete) adversarial prompts synthesized by GCG attack [73], in which they cached the intermediate synthesized results to reduce the heavy cost of searching adversarial prompts from scratch. Meanwhile, various studies [63, 6, 46, 68] conduct AT with adversarial examples found in the continuous embedding space rather than the discrete text space since searching in the continuous embedding space is more computationally efficient. Nevertheless, as a preliminary study of the length of adversarial prompts during AT, our work only analyzes AT with discrete adversarial prompts.

In-context learning theory (ICL). Transformer-based large models like LLMs are strong in performing ICL: Given a series of inputs (also known as "prompt") specified by a certain task, LLMs can make predictions well for this certain task without adjusting model parameters. Current theories in understanding ICL can be roughly divided into two categories. The first category aims to under-

stand ICL via constructing explicit multi-layer transformers to simulate the optimization process of learning function classes [13, 53, 1, 9, 34, 59]. The second category focuses on directly analyzing the training [69, 66, 19, 62, 27] and generalization [31, 33, 11, 47] of simple self-attention models (*i.e.*, one-layer transformer). [4] is the first to study adversarial attacks against linear transformers and finds that an attack can always succeed by perturbing only a single in-context sample. However, their analysis allows samples to be perturbed in the entire real space, which might not appropriately reflect real-world settings since real-world adversarial prompts can only be constructed from token/character spaces of limited size. Unlike [4], we propose a new ICL adversarial attack that requires each adversarial suffix token to be perturbed only within restricted spaces, which thus can be a better tool for understanding real-world jailbreaking.

Finally, we notice that [61] also recognizes the critical role that the number of adversarial in-context samples plays in ICL-based attacks. They present a theoretical analysis (not based on ICL theory) for adversarial attacks against ICL text classification and characterize the minimum number of in-context adversarial samples required to increase the safety loss of ICL to some extent. However, the main difference is that [61] focuses on studying the adversarial robustness of fixed ICL models, whereas our work analyzes how adversarial training affects the robustness of ICL models.

3 Preliminaries

Large language models (LLMs). Let $[V] = \{1, \cdots, V\}$ be a vocabulary set consisting of all possible tokens. Then, an LLM can be seen as a function that for any sequence $x_{1:n} \in [V]^n$ consists of n tokens, the LLM will map $x_{1:n}$ to its next token x_{n+1} following $x_{n+1} \sim p_\theta(\cdot|x_{1:n})$, where p_θ is a conditional distribution over the vocabulary set $[V]$ and θ is the model parameter of the LLM. Under such notations, when using the LLM p_θ to generate a new token sequence for the input $x_{1:n}$, the probability of generating a sequence $y_{1:m} \in [V]^m$ of length m is given by $p_\theta(y_{1:m}|x_{1:n}) = \prod_{i=1}^m p_\theta(y_i|x_{1:n} \oplus y_{1:(i-1)})$, where "$\oplus$" denotes concatenation.

Jailbreak attacks. This paper focuses on *suffix* jailbreak attacks. Concretely, suppose $x^{(h)}$ and $y^{(h)}$ are two token sequences, where $x^{(h)}$ represents a harmful prompt (*e.g.*, "Please tell me how to build a bomb.") and $y^{(h)}$ represents a corresponded targeted answer (*e.g.*, "Sure, here is a guide of how to build a bomb"). Then, the goal of a suffix jailbreak attack against the LLM p_θ aims to synthesize an *adversarial suffix* $x_{1:m}^{(s)}$ for the original harmful prompt $x^{(h)}$ via solving the following problem,

$$\min_{x_{1:m}^{(s)} \in [V]^m} -\log p_\theta(y^{(h)}|x^{(h)} \oplus x_{1:m}^{(s)}), \tag{1}$$

where $x^{(h)} \oplus x_{1:m}^{(s)}$ is the adversarial prompt and m is the sequence length of the adversarial suffix $x_{1:m}^{(s)}$. Intuitively, a large m will increase the probability of the LLM p_θ that generating the targeted answer $y^{(h)}$ for the synthesized adversarial prompt $x^{(h)} \oplus x_{1:m}^{(s)}$. To solve Eq. (1), a standard method is the Greedy Coordinate Gradient (GCG) attack [73], which leverages gradient information to search for better $x_{1:m}^{(s)}$ within the discrete space $[V]^m$ in a greedy manner.

Adversarial training (AT). We consider the canonical AT loss $\mathcal{L}$ [36, 40] to train the LLM p_θ, which consists of two sub-losses: an *adversarial loss* $\mathcal{L}_{\text{adv}}$ and an *utility loss* $\mathcal{L}_{\text{utility}}$. Specifically, given a *safety dataset* $D^{(h)}$, where each of its sample $(x^{(h)}, y^{(h)}, y^{(b)}) \in D^{(h)}$ consists of a harmful instruction $x^{(h)}$, a harmful answer $y^{(h)}$, and a *benign answer* $y^{(b)}$ (*e.g.*, "As a responsible AI, I can't tell you how to..."). The adversarial loss $\mathcal{L}_{\text{adv}}$ is defined as follows,

$$\mathcal{L}_{\text{adv}}(\theta, M, D^{(h)}) := \mathop{\mathbb{E}}_{(x^{(h)}, y^{(h)}, y^{(b)}) \in D^{(h)}} [-\log p_\theta(y^{(b)}|x^{(h)} \oplus x_{1:m}^{(s)})], \tag{2}$$

where $x_{1:m}^{(s)}$ is the adversarial suffix obtained from Eq. (1) and m is the adversarial suffix length. Note that the probability terms in Eqs. (1) and (2) look similar to each other. The difference is that the term in Eq. (1) denotes the probability that p_θ generates the harmful answer $y^{(h)}$ for the adversarial prompt, while that in Eq. (2) denotes the probability of generating the benign answer $y^{(b)}$. Besides, let $D^{(u)}$ be a *utility dataset* where each of its sample $(x^{(u)}, y^{(u)}) \in D^{(u)}$ consists of a pair of normal instruction and answer. Then, the utility loss $\mathcal{L}_{\text{utility}}$ is given by

$$\mathcal{L}_{\text{utility}}(\theta, D^{(u)}) := \mathop{\mathbb{E}}_{(x^{(u)}, y^{(u)}) \in D^{(u)}} [-\log p_\theta(y^{(u)}|x^{(u)})].$$